Let us calculate how long they travelled in one hour?

- Distance travelled by the car in one hour = ____80____ km (160/2)
- Distance travelled by the bus in one hour = _____ km
- Distance travelled by the truck in one hour = _____ km

Have you found out? Say now.

Who is fast? _____,

Who is slow? _____

Have you noticed that saying who is fast or slow is easy when we calculate the distance they travelled in one hour? In other words, you divide the distance travelled by the time taken to get the speed.

The distance travelled by an object in unit time is called speed of the object.

If an object travelled a distance 'd' in time 't' then, its speed is given as:

$$\text{Speed (s)} = \frac{\text{Distance travelled}}{\text{Time taken}} = \frac{d}{t}$$

Suppose a car travels 300 km in one hour. Then we say that the speed of the car is '300 kmph' (We read it as 'three hundred kilometres per hour').

If an object travelled 10 metre in 2 second, then its speed is given as:

Speed (s) = Distance travelled (d) / Time taken (t)

= 10 metre/ 2 second

= 5 metre / second

A bus takes three hours to cover a distance of 180 kilometres. Then its speed is given as:

Speed (s) = Distance travelled (d) / Time taken (t)

= 180 kilometre/ 3 hour

= 60 kilometre / hour

Note that metre/second or kilometre/ hour comes next to our answer for speed. What is it?

Observe the formula for speed. If we denote the distance in metre and time by second then the unit of speed is metre/second. If we denote the distance in kilometre and time in hour then the unit of speed is kilometre/ hour. Sometimes we use units like centimetre/second.

In science we generally use SI units. In SI units the unit of distance is metre and the unit of time is second. So, the SI unit of speed is metre/second.

Let us calculate

1. A car travelled 150 metre in 10 second. What is its speed?
2. Priya rides her bicycle 40 km in two hours. What is her speed?